Distributed MIS Algorithms for Rational Agents using Games

Nithin Salevemula and Shreyas Pai
Indian Institute of Technology Madras, India
nithinprakash9999@gmail.com, shreyas@cse.iitm.ac.in

1. Problem

Compute a Maximal Independent Set (MIS) in a distributed network of rational agents, where every agent prefers to be included in the MIS.

2. Model

Communication Model: LOCAL, synchronous.

Agents: Rational agents which act according to their own self-interest.

Output: Each agent irrevocably outputs one of $\{\mathbf{0}, \mathbf{1}, \perp\}$ which will be visible to neighbors once output

$1 =$ In MIS $\quad 0 =$ Not in MIS $\quad \perp =$ Abort

3. Utility Function

- **Locally Invalid MIS**: utility $= -\infty$
- **Locally Valid MIS and output** 1: utility > 0
- **Locally Valid MIS and output** 0: utility $= 0$
- **Locally Abort solution**: utility $= 0$

4. Our Algorithm

while agent is active:

- **Round 1:** Play Rock–Paper–Scissors independently with each active neighbor, selecting Rock, Paper, or Scissors uniformly at random.
- **Round 2:** If won with all active neighbors, join MIS, output 1
- **Round 3:** If any neighbor joined MIS, output 0

If any neighboring agent deviates, ignore the above algorithm and output $\perp$.

5. Properties of the Algorithm

- **Rationality Resilient:** No profitable unilateral deviation.
- **Fairness:** Every agent has a positive probability of joining the MIS.
- **Running Time:** Terminates in $O\left(3^{4\Delta} \log n\right)$ rounds with high probability

6. Future Work

The proposed algorithm is *utility-sensitive*: a small change in the agents' utility model can invalidate its rationality guarantees. In particular, if agents prefer *not* to join the MIS, the algorithm is no longer rationality-resilient. Designing MIS algorithms that remain rationality-resilient under more general utility models is an interesting direction for future work.

What is your poster about?

We're studying how to compute a Maximal Independent Set (MIS) in a distributed network of rational agents.

So what's different about your setting?

Classical algorithms assume every agent follows the protocol. In our setting, agents are rational and may deviate whenever it's in their best interest.

Does that mean the existing MIS algorithms won't work?

Yes. Since they rely on agents tossing a coin honestly, which rational agents may manipulate, or on unique IDs, which can introduce unfairness.

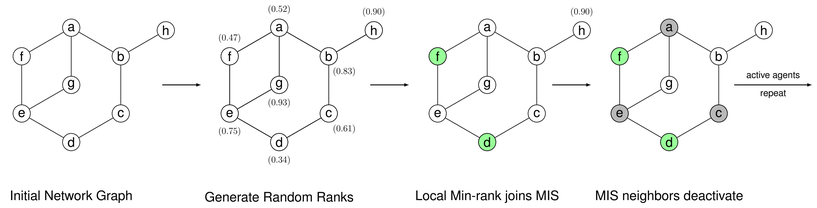
If you can't rely on those, how do you break ties?

We ask agents to play Rock–Paper–Scissors with neighbors to break ties.



Preprint: arXiv:2511.16533

Luby's in Altruistic / Indifferent Network

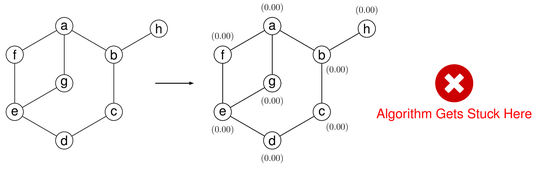


Initial Network Graph | Generate Random Ranks | Local Min-rank joins MIS | MIS neighbors deactivate

Rational Agent's Preference

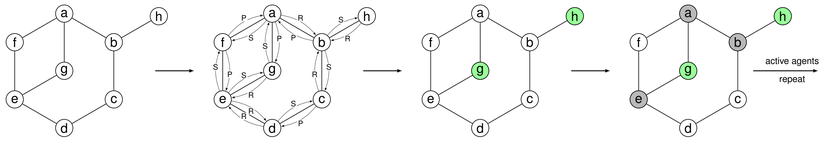
Valid MIS and In MIS $\succ$ Valid MIS and Not In MIS $\succ$ Invalid MIS

Luby's in rational network with above utility function



Every agents claim Rank 0.00

Our Algorithm



Exchange RPS | agents winning all RPS games join MIS

Future work

Valid MIS and Not In MIS $\succ$ Valid MIS and In MIS $\succ$ Invalid MIS

Thank you for visiting our poster!

Comments, questions, and suggestions are greatly appreciated.